

OS Skill 1

```
mahid@Mahidhar:~$ sudo apt update
[sudo: authenticat] Password:
Get:1 http://archive.ubuntu.com/ubuntu resolute InRelease [136 kB]
Get:2 http://security.ubuntu.com/ubuntu resolute-security InRelease [137 kB]
Get:3 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Packages [417 kB]
Get:4 http://security.ubuntu.com/ubuntu resolute-security/main Translation-en [97.2 kB]
Get:5 http://security.ubuntu.com/ubuntu resolute-security/main amd64 Components [46.7 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Get:7 http://security.ubuntu.com/ubuntu resolute-security/main amd64 c-n-f Metadata [4616 B]
Get:8 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Packages [159 kB]
Get:9 http://security.ubuntu.com/ubuntu resolute-security/universe Translation-en [51.2 kB]
Get:10 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 Components [43.2 kB]
Get:11 http://security.ubuntu.com/ubuntu resolute-security/universe amd64 c-n-f Metadata [3552 B]
Get:12 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 Packages [349 kB]
Get:13 http://security.ubuntu.com/ubuntu resolute-security/restricted Translation-en [67.8 kB]
Get:14 http://security.ubuntu.com/ubuntu resolute-security/restricted amd64 c-n-f Metadata [396 B]
Get:15 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Packages [9820 B]
Get:16 http://security.ubuntu.com/ubuntu resolute-security/multiverse Translation-en [2660 B]
Get:17 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 Components [212 B]
Get:18 http://security.ubuntu.com/ubuntu resolute-security/multiverse amd64 c-n-f Metadata [120 B]
Get:19 http://archive.ubuntu.com/ubuntu resolute-backports InRelease [136 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute/main amd64 Packages [1480 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute/main Translation-en [524 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute/main amd64 Components [395 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute/main amd64 c-n-f Metadata [32.4 kB]
Get:24 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Packages [16.0 MB]
Get:25 http://archive.ubuntu.com/ubuntu resolute/universe Translation-en [6329 kB]
Get:26 http://archive.ubuntu.com/ubuntu resolute/universe amd64 Components [4556 kB]
Get:27 http://archive.ubuntu.com/ubuntu resolute/universe amd64 c-n-f Metadata [313 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Packages [152 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute/restricted Translation-en [25.8 kB]
Get:30 http://archive.ubuntu.com/ubuntu resolute/restricted amd64 Components [556 B]
Get:31 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Packages [290 kB]
Get:32 http://archive.ubuntu.com/ubuntu resolute/multiverse Translation-en [127 kB]
Get:33 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 Components [50.0 kB]
Get:34 http://archive.ubuntu.com/ubuntu resolute/multiverse amd64 c-n-f Metadata [8276 B]
Get:35 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [492 kB]
Get:36 http://archive.ubuntu.com/ubuntu resolute-updates/main Translation-en [118 kB]
Get:37 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 Components [97.8 kB]
Get:38 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 c-n-f Metadata [5756 B]

Get:39 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [243 kB]
Get:40 http://archive.ubuntu.com/ubuntu resolute-updates/universe Translation-en [81.0 kB]
Get:41 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Components [184 kB]
Get:42 http://archive.ubuntu.com/ubuntu resolute-updates/universe amd64 c-n-f Metadata [4700 B]
Get:43 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 Packages [367 kB]
Get:44 http://archive.ubuntu.com/ubuntu resolute-updates/restricted Translation-en [70.8 kB]
Get:45 http://archive.ubuntu.com/ubuntu resolute-updates/restricted amd64 c-n-f Metadata [392 B]
Get:46 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Packages [11.3 kB]
Get:47 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse Translation-en [3032 B]
Get:48 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 Components [216 B]
Get:49 http://archive.ubuntu.com/ubuntu resolute-updates/multiverse amd64 c-n-f Metadata [256 B]
Get:50 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 Components [212 B]
Get:51 http://archive.ubuntu.com/ubuntu resolute-backports/main amd64 c-n-f Metadata [112 B]
Get:52 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Packages [640 B]
Get:53 http://archive.ubuntu.com/ubuntu resolute-backports/universe Translation-en [300 B]
Get:54 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 Components [216 B]
Get:55 http://archive.ubuntu.com/ubuntu resolute-backports/universe amd64 c-n-f Metadata [116 B]
Get:56 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 Components [216 B]
Get:57 http://archive.ubuntu.com/ubuntu resolute-backports/restricted amd64 c-n-f Metadata [120 B]
Get:58 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 Components [216 B]
Get:59 http://archive.ubuntu.com/ubuntu resolute-backports/multiverse amd64 c-n-f Metadata [120 B]
Fetched 33.7 MB in 7s (5094 kB/s)
119 packages can be upgraded. Run 'apt list --upgradable' to see them.
mahid@Mahidhar:~$ sudo apt install gcc git make valgrind -y
Error: Unable to locate package valgrind
mahid@Mahidhar:~$ sudo apt install gcc git make valgrind -y
git is already the newest version (1:2.53.0-1ubuntu1).
git set to manually installed.
Upgrading:
  libc-bin  libc-gconv-modules-extra  libc6  locales

Installing:
  gcc  make  valgrind

Installing dependencies:
  cpp          gcc-15-x86-64-linux-gnu  libc6-dbg          libgcc-15-dev  liblsan0          libubsan1
  cpp-15       gcc-x86-64-linux-gnu    libc6-dev          libgomp1       libmpc3           linux-libc-dev
  cpp-15-x86-64-linux-gnu  gdb                  libc1-0            libhwasan0     libquadmath0     manpages-dev
  cpp-x86-64-linux-gnu      libasan8             libdebuginfod-common  libipt2        libsource-highlight-common  rpcsvc-proto
  gcc-15              libbabeltrace1       libdebuginfod1t64    libisl23       libsource-highlight4t64
```

```
Suggested packages:
  cpp-doc      cpp-15-doc    autoconf     libtool      bison        gcc-15-multilib  gdb-x86-64-linux-gnu  gdbserver    glibc-doc    valgrind-mpi
  gcc-15-locale  gcc-multilib  automake     flex         gcc-doc      gcc-15-doc       gdb-doc               libc-devtools  make-doc     kcache-grind
```

Summary:

```
Upgrading: 4, Installing: 37, Removing: 0, Not Upgrading: 115
Download size: 95.2 MB
Space needed: 312 MB / 1024 GB available
```

```
Get:1 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-gconv-modules-extra amd64 2.43-2ubuntu2.3 [1362 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc6 amd64 2.43-2ubuntu2.3 [2103 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libc-bin amd64 2.43-2ubuntu2.3 [701 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod-common all 0.194-4 [12.3 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 locales all 2.43-2ubuntu2.3 [4225 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15-base amd64 15.2.0-16ubuntu1 [39.0 kB]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libisl23 amd64 0.27-1build1 [691 kB]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmpc3 amd64 1.3.1-3 [54.6 kB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 cpp-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [12.9 MB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 cpp-15 amd64 15.2.0-16ubuntu1 [1030 B]
Get:11 http://archive.ubuntu.com/ubuntu resolute/main amd64 cpp-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [5840 B]
Get:12 http://archive.ubuntu.com/ubuntu resolute/main amd64 cpp amd64 4:15.2.0-5ubuntu1 [22.4 kB]
Get:13 http://archive.ubuntu.com/ubuntu resolute/main amd64 libcc1-0 amd64 16-20260322-1ubuntu1 [51.2 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main amd64 libgomp1 amd64 16-20260322-1ubuntu1 [162 kB]
Get:15 http://archive.ubuntu.com/ubuntu resolute/main amd64 libitm1 amd64 16-20260322-1ubuntu1 [30.0 kB]
Get:16 http://archive.ubuntu.com/ubuntu resolute/main amd64 libasan8 amd64 16-20260322-1ubuntu1 [3184 kB]
Get:17 http://archive.ubuntu.com/ubuntu resolute/main amd64 liblsan0 amd64 16-20260322-1ubuntu1 [1394 kB]
Get:18 http://archive.ubuntu.com/ubuntu resolute/main amd64 libtsan2 amd64 16-20260322-1ubuntu1 [2849 kB]
Get:19 http://archive.ubuntu.com/ubuntu resolute/main amd64 libubsan1 amd64 16-20260322-1ubuntu1 [1240 kB]
Get:20 http://archive.ubuntu.com/ubuntu resolute/main amd64 libhwcasan0 amd64 16-20260322-1ubuntu1 [1731 kB]
Get:21 http://archive.ubuntu.com/ubuntu resolute/main amd64 libquadmath0 amd64 16-20260322-1ubuntu1 [155 kB]
Get:22 http://archive.ubuntu.com/ubuntu resolute/main amd64 libgcc-15-dev amd64 15.2.0-16ubuntu1 [2866 kB]
Get:23 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15-x86-64-linux-gnu amd64 15.2.0-16ubuntu1 [25.4 MB]
Get:24 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-15 amd64 15.2.0-16ubuntu1 [539 kB]
Get:25 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc-x86-64-linux-gnu amd64 4:15.2.0-5ubuntu1 [1208 B]
Get:26 http://archive.ubuntu.com/ubuntu resolute/main amd64 gcc amd64 4:15.2.0-5ubuntu1 [5010 B]
Get:27 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdw1t64 amd64 0.194-4 [285 kB]
Get:28 http://archive.ubuntu.com/ubuntu resolute/main amd64 libbabeltrace1 amd64 1.5.11-5build1 [165 kB]
Get:29 http://archive.ubuntu.com/ubuntu resolute/main amd64 libdebuginfod1t64 amd64 0.194-4 [20.7 kB]
Get:30 http://archive.ubuntu.com/ubuntu resolute/main amd64 libipt2 amd64 2.1.2-3 [49.2 kB]
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Get:41 http://archive.ubuntu.com/ubuntu resolute/main amd64 valgrind amd64 1:3.26.0-0ubuntu1 [15.7 MB]
```

Fetched 95.2 MB in 23s (4118 kB/s)

Extracting templates from packages: 100%

Preconfiguring packages ...

(Reading database ... 35923 files and directories currently installed.)

Preparing to unpack .../libc-gconv-modules-extra_2.43-2ubuntu2.3_amd64.deb ...

Unpacking libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...

Setting up libc-gconv-modules-extra:amd64 (2.43-2ubuntu2.3) ...

(Reading database ... 35923 files and directories currently installed.)

Preparing to unpack .../libc6_2.43-2ubuntu2.3_amd64.deb ...

Unpacking libc6:amd64 (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...

Setting up libc6:amd64 (2.43-2ubuntu2.3) ...

(Reading database ... 35923 files and directories currently installed.)

Preparing to unpack .../libc-bin_2.43-2ubuntu2.3_amd64.deb ...

Unpacking libc-bin (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...

Setting up libc-bin (2.43-2ubuntu2.3) ...

Selecting previously unselected package libdebuginfod-common.

(Reading database ... 35923 files and directories currently installed.)

Preparing to unpack .../00-libdebuginfod-common_0.194-4_all.deb ...

Unpacking libdebuginfod-common (0.194-4) ...

Preparing to unpack .../01-locales_2.43-2ubuntu2.3_all.deb ...

Unpacking locales (2.43-2ubuntu2.3) over (2.43-2ubuntu2) ...

Selecting previously unselected package gcc-15-base:amd64.

Preparing to unpack .../02-gcc-15-base_15.2.0-16ubuntu1_amd64.deb ...

Unpacking gcc-15-base:amd64 (15.2.0-16ubuntu1) ...

Selecting previously unselected package libisl23:amd64.

Preparing to unpack .../03-libisl23_0.27-1build1_amd64.deb ...

Unpacking libisl23:amd64 (0.27-1build1) ...

Selecting previously unselected package libmpc3:amd64.

Preparing to unpack .../04-libmpc3_1.3.1-3_amd64.deb ...

Unpacking libmpc3:amd64 (1.3.1-3) ...

Selecting previously unselected package cpp-15-x86-64-linux-gnu.

Preparing to unpack .../05-cpp-15-x86-64-linux-gnu_15.2.0-16ubuntu1_amd64.deb ...

Unpacking cpp-15-x86-64-linux-gnu (15.2.0-16ubuntu1) ...

Selecting previously unselected package cpp-15.

Preparing to unpack .../06-cpp-15_15.2.0-16ubuntu1_amd64.deb ...

Unpacking cpp-15 (15.2.0-16ubuntu1) ...

Selecting previously unselected package cpp-x86-64-linux-gnu.

Preparing to unpack .../07-cpp-x86-64-linux-gnu_4%3a15.2.0-5ubuntu1_amd64.deb ...

Unpacking cpp-x86-64-linux-gnu (4:15.2.0-5ubuntu1) ...

```
Selecting previously unselected package cpp.
Preparing to unpack .../08-cpp_4%3a15.2.0-5ubuntu1_amd64.deb ...
Unpacking cpp (4:15.2.0-5ubuntu1) ...
Selecting previously unselected package libcc1-0:amd64.
Preparing to unpack .../09-libcc1-0_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libcc1-0:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libgomp1:amd64.
Preparing to unpack .../10-libgomp1_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libgomp1:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libitm1:amd64.
Preparing to unpack .../11-libitm1_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libitm1:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libasan8:amd64.
Preparing to unpack .../12-libasan8_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libasan8:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package liblsan0:amd64.
Preparing to unpack .../13-liblsan0_16-20260322-1ubuntu1_amd64.deb ...
Unpacking liblsan0:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libtsan2:amd64.
Preparing to unpack .../14-libtsan2_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libtsan2:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libubsan1:amd64.
Preparing to unpack .../15-libubsan1_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libubsan1:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libhwasan0:amd64.
Preparing to unpack .../16-libhwasan0_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libhwasan0:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libquadmath0:amd64.
Preparing to unpack .../17-libquadmath0_16-20260322-1ubuntu1_amd64.deb ...
Unpacking libquadmath0:amd64 (16-20260322-1ubuntu1) ...
Selecting previously unselected package libgcc-15-dev:amd64.
Preparing to unpack .../18-libgcc-15-dev_15.2.0-16ubuntu1_amd64.deb ...
Unpacking libgcc-15-dev:amd64 (15.2.0-16ubuntu1) ...
Selecting previously unselected package gcc-15-x86-64-linux-gnu.
Preparing to unpack .../19-gcc-15-x86-64-linux-gnu_15.2.0-16ubuntu1_amd64.deb ...
Unpacking gcc-15-x86-64-linux-gnu (15.2.0-16ubuntu1) ...
Selecting previously unselected package gcc-15.
Preparing to unpack .../20-gcc-15_15.2.0-16ubuntu1_amd64.deb ...
Unpacking gcc-15 (15.2.0-16ubuntu1) ...
Selecting previously unselected package gcc-x86-64-linux-gnu.
```

```
mahid@Mahidhar:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
Copyright (C) 2025 Free Software Foundation, Inc.
This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

mahid@Mahidhar:~$ git --version
git version 2.53.0
mahid@Mahidhar:~$ valgrind --vmesion
valgrind: no program specified
valgrind: Use --help for more information.
mahid@Mahidhar:~$ mkdir ossp
mahid@Mahidhar:~$ cd ossp
mahid@Mahidhar:~/ossp$ mkdir src include build
mahid@Mahidhar:~/ossp$ ls
build  include  src
mahid@Mahidhar:~/ossp$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/mahid/ossp/.git/
mahid@Mahidhar:~/ossp$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
mahid@Mahidhar:~/ossp$ touch READ.me
mahid@Mahidhar:~/ossp$ git add
Nothing specified, nothing added.
```



```

Nothing specified, nothing added.
hint: Maybe you wanted to say 'git add .'
hint: Disable this message with "git config set advice.addEmptyPaths false"
mahid@Mahidhar:~/ossp$ git commit -m "Initial initial project setup"
Author identity unknown

*** Please tell me who you are.

Run

git config --global user.email "you@example.com"
git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <mahid@Mahidhar.localdomain>) not allowed
mahid@Mahidhar:~/ossp$ cs src
Command 'cs' not found, but can be installed with:
sudo apt install csound
mahid@Mahidhar:~/ossp$ nano.c
Command 'nano.c' not found, did you mean:
  command 'nanoc' from deb nanoc (4.14.6-1)
Try: sudo apt install <deb name>
mahid@Mahidhar:~/ossp$ nano main.c
mahid@Mahidhar:~/ossp$ gcc main.c -o ../build/main
/usr/bin/x86_64-linux-gnu-ld.bfd: cannot open output file ../build/main: No such file or directory
collect2: error: ld returned 1 exit status
mahid@Mahidhar:~/ossp$ ../build/main
-bash: ../build/main: No such file or directory
mahid@Mahidhar:~/ossp$ nano fork_demo.c
mahid@Mahidhar:~/ossp$ gcc firk_demo.c -o fork_demo
cc1: fatal error: firk_demo.c: No such file or directory
compilation terminated.
mahid@Mahidhar:~/ossp$ gcc fork_demo.c -o fork_demo
mahid@Mahidhar:~/ossp$ ./fork_demo
Parent Process
Parent PID: 2425
Child Process
Child PID 2426

Child PID 2426
mahid@Mahidhar:~/ossp$ nano exec_demo.c
mahid@Mahidhar:~/ossp$ gcc exec_demo.c -o exec_demo
mahid@Mahidhar:~/ossp$ ./exec_demo
Before exec
total 56
-rw-r--r-- 1 mahid mahid    0 Aug 24 05:47 READ.me
drwxr-xr-x 2 mahid mahid  4096 Aug 24 05:46 build
-rwxr-xr-x 1 mahid mahid 16000 Aug 24 06:43 exec_demo
-rw-r--r-- 1 mahid mahid   166 Aug 24 06:43 exec_demo.c
-rwxr-xr-x 1 mahid mahid 16080 Aug 24 06:39 fork_demo
-rw-r--r-- 1 mahid mahid   287 Aug 24 06:37 fork_demo.c
drwxr-xr-x 2 mahid mahid  4096 Aug 24 05:46 include
-rw-r--r-- 1 mahid mahid    70 Aug 24 05:51 main.c
drwxr-xr-x 2 mahid mahid  4096 Aug 24 05:46 src
mahid@Mahidhar:~/ossp$ |

```